
Strings in AdS_3 - $\text{SL}(2, \mathbb{R})$ WZW Model

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ABSTRACT

This work presents a systematic study of bosonic string theory on a 26-dimensional background of the form $\text{AdS}_3 \times \mathcal{M}$, where $\mathcal{M}$ is a compact Euclidean manifold, within the framework of the $\text{SL}(2, \mathbb{R})$ Wess–Zumino–Witten (WZW) model. After reviewing the realization of AdS_3 as the group manifold of $\text{SL}(2, \mathbb{R})$, we derive the equations of motion, conserved currents, Virasoro constraints, and classical geodesic solutions of the theory. Particular emphasis is placed on the spectral flow, which is shown to transform geodesic solutions into short- and long-string configurations. This provides a natural physical interpretation of winding sectors and highlights their role in the string spectrum on AdS_3 .

The quantum theory is developed through the construction of the affine $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ current (Kac–Moody) algebra and the associated Virasoro algebra via the Sugawara construction. The representation theory of both the global and affine $\text{SL}(2, \mathbb{R})$ algebras is investigated, including unitary discrete and continuous representations, their wavefunctions on AdS_3 , and the no-ghost theorem. It is shown that unflowed representations alone are insufficient to describe the physical spectrum and that the inclusion of spectrally flowed sectors is essential for a consistent description of both short- and long-string states.

To further investigate the structure of the theory, the model is analytically continued to the Euclidean H_3^+ WZW model. Using bosonization and parafermionic techniques, primary fields and long-string scattering amplitudes are studied in a free-field framework. The analytic structure of the reflection amplitude is shown to encode information about normalizable discrete states and the physical spectrum. In particular, the No-Ghost bound $\frac{1}{2} < j < \frac{k-1}{2}$ is recovered from the pole structure of the reflection coefficient, establishing a direct connection between long-string scattering and affine representation.

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1 Introduction

String theory in curved spacetime is notoriously difficult because the associated worldsheet theory is typically a nonlinear sigma model for which exact quantization is rarely possible. So, much of our understanding relies on low-energy effective descriptions or perturbative expansions around weakly curved backgrounds. A notable exception is the AdS_3 space supported by the NS flux, where the worldsheet theory admits an exact description in terms of the $\text{SL}(2, \mathbb{R})$ WZW model. This provides a unique setting in which many aspects of string dynamics can be studied beyond the super-gravity approximation.

The AdS_3 background occupies a distinguished position in string theory. Besides serving as one of the simplest negatively curved spacetimes, it possesses an infinite-dimensional symmetry structure arising from affine current algebras on the worldsheet. These symmetries provide an exact worldsheet description of the theory and make it possible to study the physical string spectrum in considerable detail. At the same time, the non-compact nature of $\text{SL}(2, \mathbb{R})$ introduces several conceptual challenges that are absent in compact-group WZW models. In particular, the appearance of negative-norm states, the existence of long-string configurations, and the role of winding sectors require a careful treatment of the representation theory of the affine current algebra.

The primary objective of this work is to understand how the spectrum of string theory in AdS_3 emerges from the $\text{SL}(2, \mathbb{R})$ WZW model. To this end, we begin with the classical geometry of AdS_3 and its realization as a group manifold. We derive the WZW action, investigate its chiral symmetries, and construct the associated current algebra. Classical string solutions are then studied through geodesics and monodromy classes, providing an intuitive understanding of short- and long-string configurations. This analysis naturally leads to the notion of spectral flow, which plays a central role in the structure of the theory.

The second part of the work focuses on quantization. We develop the affine Kac–Moody and the Virasoro algebra (Sugawara construction) of the theory and study the representation content of the Hilbert space. We examine both discrete and continuous representations of $\text{SL}(2, \mathbb{R})$, discuss the no-ghost theorem, and show why the traditional unflowed representations are insufficient to describe the full physical spectrum. Spectral-flowed sectors are then introduced and interpreted as the worldsheet realization of winding string configurations. Their inclusion is essential for obtaining a consistent spectrum and naturally incorporates the long-string states associated with continuous representations.

Finally, we study long-string dynamics through scattering theory. After Wick rotating to the Euclidean H_3^+ WZW model, we employ bosonization and parafermionic techniques to study primary fields and reflection amplitudes, relating the physical spectrum to the analytic structure of correlation functions. Together, these results provide a unified picture of string propagation in AdS_3 , illustrating how spectral flow, affine representation theory, and long-string dynamics combine to produce a consistent ghost-free spectrum.

2 Classical Structure of Strings in AdS₃

2.1 AdS₃ as an SL(2, ℝ) Group Manifold

The group SL(2, ℝ) consists of all real 2×2 matrices with unit determinant. A general group element can be parameterized as,

$$g = \begin{pmatrix} X_{-1} + X_1 & X_0 - X_2 \\ -X_0 - X_2 & X_{-1} - X_1 \end{pmatrix}, \quad \det(g) = 1 \implies -X_{-1}^2 - X_0^2 + X_1^2 + X_2^2 = -1 \quad (1)$$

This equation describes a three-dimensional hyperboloid embedded in the flat space $\mathbb{R}^{2,2}$ with metric $ds_{\mathbb{R}^{2,2}}^2 = -dX_{-1}^2 - dX_0^2 + dX_1^2 + dX_2^2$. We can parameterize the embedded manifold using the intrinsic-global coordinates (t, ρ, ϕ) through

$$\begin{aligned} X_{-1} &= \cosh(\rho) \cos(t), & X_0 &= \cosh(\rho) \sin(t), \\ X_1 &= \sinh(\rho) \cos(\phi), & X_2 &= \sinh(\rho) \sin(\phi) \end{aligned} \quad (2)$$

Then the induced metric on the hyperboloid becomes precisely the metric of global AdS₃.

$$ds_{\text{AdS}_3}^2 = \cosh^2(\rho) dt^2 + d\rho^2 + \sinh^2(\rho) d\phi^2 \quad (3)$$

It can be shown that the induced metric on AdS₃ can be expressed as

$$ds^2 = \frac{1}{2} \text{Tr}(g^{-1} dg g^{-1} dg), \quad (4)$$

where

$$g = \begin{pmatrix} \cosh(\rho) \cos(t) + \sinh(\rho) \cos(\phi) & \cosh(\rho) \sin(t) - \sinh(\rho) \sin(\phi) \\ -\cosh(\rho) \sin(t) - \sinh(\rho) \sin(\phi) & \cosh(\rho) \cos(t) - \sinh(\rho) \cos(\phi) \end{pmatrix} = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2} \quad (5)$$

$$\text{with} \quad u = \frac{t + \phi}{2}, \quad v = \frac{t - \phi}{2}.$$

This identification of AdS₃ as the group manifold of SL(2, ℝ) has important consequences. The spacetime inherits left- and right-acting isometries $g \longrightarrow g_L g g_R^{-1}$ which generate two independent copies of the global $\text{SL}(2, \mathbb{R})_L \times \text{SL}(2, \mathbb{R})_R$ symmetry. In the WZW description, these symmetries are promoted to left- and right-moving affine current algebras on the worldsheet, providing the algebraic framework underlying the string spectrum.

Because the geometry itself is encoded in a group manifold, string propagation in AdS₃ can be described exactly by an SL(2, ℝ) WZW model. This exact description distinguishes AdS₃ from most curved string backgrounds and provides a powerful framework for studying both classical and quantum aspects of the theory.

2.2 The $\text{SL}(2, \mathbb{R})$ WZW Model and Chiral Symmetries

Having identified AdS_3 with the group manifold of $\text{SL}(2, \mathbb{R})$, string propagation on this background is described by an exact WZW model whose dynamical field is a map from the two-dimensional worldsheet Σ into the target-space group manifold as shown below.

$$g(\tau, \sigma) : \Sigma \rightarrow \text{SL}(2, \mathbb{R}) \quad \implies \quad g \in \text{SL}(2, \mathbb{R}) \quad (6)$$

Then, the WZW action is given by,

$$S = \underbrace{\frac{-k}{16\pi} \int_{\Sigma} d^2\sigma \sqrt{-h} h^{\alpha\beta} \text{Tr}(g^{-1}(\partial_{\alpha}g) g^{-1}(\partial_{\beta}g))}_{S_{\text{kin}}} + \underbrace{\frac{k}{24\pi} \int_B \text{Tr}(g^{-1}dg \wedge g^{-1}dg \wedge g^{-1}dg)}_{\Gamma_{\text{WZ}}} \quad (7)$$

where B is a three-dimensional manifold such that $\partial B = \Sigma$ and $h_{\alpha\beta}$ is the world-sheet metric. The first term in the WZW action is known as the *sigma-model kinetic term*. The second term, known as the Wess–Zumino term, is topological as it is independent of the world-sheet metric. It describes the coupling of string to the background antisymmetric B -field or, equivalently, to the NS–NS three-form flux supporting the AdS_3 background.

The left ($\delta g = \varepsilon g$) and right ($\delta g = g\varepsilon$) variations of the WZW action, with $\varepsilon \in \mathfrak{sl}(2, \mathbb{R})$, yield the equations of motion and imply the existence of the $\mathfrak{sl}(2, \mathbb{R})$ Lie algebra valued conserved chiral currents as follows,

$$J_L(x^-) = \frac{k}{2} g^{-1}(\partial_- g) ; \partial_+ (J_L(x^-)) = 0 \quad \text{and} \quad J_R(x^+) = \frac{k}{2} (\partial_+ g) g^{-1} ; \partial_- (J_R(x^+)) = 0 \quad (8)$$

Thus, the left-moving current J_L depends only on the left-moving coordinate $x^- = \tau - \sigma$, while the right-moving current depends only on the right-moving coordinate $x^+ = \tau + \sigma$.

The WZW action possesses a local $\text{SL}(2, \mathbb{R}) \times \text{SL}(2, \mathbb{R})$ symmetry under the left and right transformations $g(x^+, x^-) \longrightarrow \Omega_L(x^-) g(x^+, x^-) \Omega_R^{-1}(x^+)$ where $\Omega_L, \Omega_R \in \text{SL}(2, \mathbb{R})$. The conserved currents associated with this symmetry generate two independent copies of the $\mathfrak{sl}(2, \mathbb{R})$ Lie algebra. Upon quantization, these become affine Kac–Moody current algebras, which constitute the fundamental symmetry structure of the worldsheet theory.

2.3 Chiral Factorization, Conserved Charges and currents

An important consequence of the equations of motion is that classical solutions factor into left and right sectors as follows.

$$g(x^+, x^-) = g_+(x^+) g_-(x^-) \quad (9)$$

This decomposition separates the left- and right-moving degrees of freedom and forms the basis for the analysis of classical string solutions.

For closed strings, the worldsheet coordinate $\sigma \sim \sigma + 2\pi$ is periodic, and therefore the full field must satisfy the periodicity condition $g(\tau, \sigma + 2\pi) = g(\tau, \sigma)$.

$$\begin{aligned} \implies g_+(x^+ + 2\pi) g_-(x^- - 2\pi) &= g_+(x^+) g_-(x^-) \\ \implies g_+(x^+)^{-1} g_+(x^+ + 2\pi) &= g_-(x^-) g_-(x^- - 2\pi)^{-1} = M \quad (\text{Monodromy Matrix}) \end{aligned} \quad (10)$$

Thus, the chiral factors are periodic up to multiplication by a constant group element called the Monodromy matrix M , which measures their failure to be individually periodic.

$$\implies g_+(x^+ + 2\pi) = g_+(x^+) M \quad \text{and} \quad g_-(x^- - 2\pi) = M^{-1} g_-(x^-) \quad (11)$$

The decomposition $g = g_+ g_-$ is not unique, as $g_+ \rightarrow g_+ h$ and $g_- \rightarrow h^{-1} g_-$ with constant $h \in \text{SL}(2, \mathbb{R})$ leaves g invariant, under which $M \rightarrow h^{-1} M h$. Hence, the monodromy matrix is defined only up to conjugation, and its physical content is encoded in its conjugacy class. Classical string solutions are accordingly classified into three sectors:

- Elliptic Monodromy $\rightarrow$ short strings oscillating near the center of AdS_3 .
- Hyperbolic Monodromy $\rightarrow$ long strings reaching toward the boundary.
- Parabolic Monodromy $\rightarrow$ threshold configurations separating the two sectors.

Since the conserved currents are Lie algebra valued, we can expand them in terms of the generators of $\mathfrak{sl}(2, \mathbb{R})$. The 2×2 matrix generators of $\mathfrak{sl}(2, \mathbb{R})$ are,

$$\begin{aligned} T^1 &= \frac{1}{2} \sigma_3, \quad T^2 = \frac{1}{2} \sigma_1, \quad T^3 = -\frac{i}{2} \sigma_2 \quad \text{and} \quad T^\pm = T^1 \pm iT^2 \\ \implies [iT^3, T^\pm] &= \pm T^\pm, \quad [T^+, T^-] = 2iT^3 \end{aligned} \quad (12)$$

We can also choose another set of bases defined by a $\mathbb{Z}_2$ transformation denoted by T^{a*} , which are given by $T^{\pm*} = T^\mp$ and $T^{3*} = T^3$. They also follow the same commutation relation as before. The invariant symmetric bilinear killing form κ^{ab} of $\mathfrak{sl}(2, \mathbb{R})$, which is used to raise and lower indices, is defined by

$$\text{Tr}(T^a T^b) = \text{Tr}(T^{a*} T^{b*}) = \frac{\kappa^{ab}}{2} \quad \text{where} \quad \kappa^{ab} = \text{diag}(1, 1, -1) = \kappa_{ab} \quad (13)$$

The chiral currents can be expanded in terms of the basis as follows.

$$\begin{aligned} J_R(x^+) &= \frac{k}{2} (\partial_+ g) g^{-1} = \sum_{a,b} \kappa_{ab} J_R^a(x^+) T^b \implies J_R^a(x^+) = k \text{Tr}(T^a (\partial_+ g) g^{-1}) \\ J_L(x^-) &= \frac{k}{2} g^{-1} (\partial_- g) = \sum_{a,b} J_L^a(x^-) T^{b*} \kappa_{ab} \implies J_L^a(x^-) = k \text{Tr}(T^{a*} g^{-1} (\partial_- g)) \end{aligned} \quad (14)$$

Using the parameterization of the $\text{SL}(2, \mathbb{R})$ group element as $g = e^{iu\sigma_2} e^{\rho\sigma_3} e^{iv\sigma_2}$, where $u = \frac{1}{2}(t + \phi)$ and $v = \frac{1}{2}(t - \phi)$, we get,

$$\begin{aligned} J_R^3 &= k (\cosh(2\rho) (\partial_+ v) + (\partial_+ u)) \quad \text{and} \quad J_R^\pm = k \left((\partial_+ \rho) \pm i(\partial_+ v) \sinh(2\rho) \right) e^{\mp 2iu} \\ J_L^3 &= k (\cosh(2\rho) (\partial_- u) + (\partial_- v)) \quad \text{and} \quad J_L^\pm = k \left((\partial_- \rho) \pm i(\partial_- u) \sinh(2\rho) \right) e^{\mp 2iv} \end{aligned} \quad (15)$$

We can also relate the zero modes of $J_{L,R}^3$ conserved currents with the space-time energy and angular momentum. The global time translation and angular rotation symmetries of AdS_3 arise from the below left and right $\text{SL}(2, \mathbb{R})$ actions on the group element g .

$$g \xrightarrow[t \rightarrow t + \delta t]{\quad} e^{-T^3 \delta t} g e^{-T^3 \delta t} \quad \text{and} \quad g \xrightarrow[\phi \rightarrow \phi + \delta \phi]{\quad} e^{-T^3 \delta \phi} g e^{T^3 \delta \phi} \quad (16)$$

Consequently, the corresponding conserved spacetime (Noether) charges are given by the zero modes of the associated worldsheet currents.

$$E = J_{0,L}^3 + J_{0,R}^3, \quad \ell = J_{0,R}^3 - J_{0,L}^3, \quad (17)$$

where

$$J_{0,L}^3 = \frac{1}{2\pi} \int_0^{2\pi} dx^- J_L^3(x^-), \quad J_{0,R}^3 = \frac{1}{2\pi} \int_0^{2\pi} dx^+ J_R^3(x^+). \quad (18)$$

Thus, the zero modes of the $J_{L,R}^3$ currents encode the AdS_3 energy and angular momentum of the string configuration. The global Killing vectors ∂_t and ∂_ϕ correspond to specific linear combinations of the left and right $\text{SL}(2, \mathbb{R})$ generators, leading to the above identification of the spacetime charges with the current zero modes.

2.4 Classical Energy - Momentum Tensor & Virasoro Constraints

Varying the complete WZW action with respect to the worldsheet metric $h_{\alpha\beta}$ (Lorentzian) defines the Energy-Momentum tensor as follows.

$$T_{\alpha\beta} = -\frac{4\pi}{\sqrt{-h}} \frac{\delta S}{\delta h^{\alpha\beta}}. \quad (19)$$

The Wess-Zumino term Γ_{WZ} is topological and does not depend on $h_{\alpha\beta}$. Hence, it does not contribute to the stress tensor. Varying the kinetic term with respect to $h^{\alpha\beta}$, then going to the light-cone coordinates, and then expanding the resulting conserved currents in terms of the $\text{SL}(2, \mathbb{R})$ generators, T^a and T^{a*} , gives,

$$T_{++}(x^+) = \frac{1}{k} \sum_{a,b} \kappa_{ab} J_R^a(x^+) J_R^b(x^+) \quad \text{and} \quad T_{--}(x^-) = \frac{1}{k} \sum_{a,b} \kappa_{ab} J_L^a(x^-) J_L^b(x^-) \quad (20)$$

Aside: The normalization constant in front of the energy-momentum tensor components is irrelevant classically, as the equations of motion, the constraint algebra, and the physical solutions remain unchanged. But in quantum theory, it is significant, as will be seen later.

We consider string propagation in a background of the form $\text{AdS}_3 \times \mathcal{M}$, where the AdS_3 part is described by an $\text{SL}(2, \mathbb{R})$ WZW model and $\mathcal{M}$ is some internal conformal field theory. Then the total worldsheet action is a sum of two decoupled sigma models:

$$S_{\text{total}} = \underbrace{S[g, h_{\alpha\beta}]}_{\text{WZW}} + S_{\mathcal{M}}[\mathcal{X}^i, h_{\alpha\beta}] \quad (21)$$

where $g(\tau, \sigma) \in \text{SL}(2, \mathbb{R})$ describes motion in AdS_3 , $\mathcal{X}^i(\tau, \sigma)$ are coordinates on $\mathcal{M}$, and $h_{\alpha\beta}$ is the worldsheet metric. Since the action splits into two independent pieces, the worldsheet stress-energy tensor also splits as $T_{\alpha\beta}^{\text{total}} = T_{\alpha\beta} + T_{\alpha\beta}^{\mathcal{M}}$, where $T_{\alpha\beta}^{\mathcal{M}}$ is left unspecified. The Polyakov string action is invariant under Worldsheet diffeomorphisms and Weyl transformations. Because of this gauge symmetry, the worldsheet metric $h_{\alpha\beta}$ has no propagating degrees of freedom; it is a non-physical gauge field. So, the equation of motion obtained by varying the action with respect to $h_{\alpha\beta}$ is $\frac{\delta S}{\delta h^{\alpha\beta}} = 0$. Using the definition of the stress tensor, we get the classical Virasoro constraint equations as,

$$T_{\alpha\beta}^{\text{total}} = T_{\alpha\beta} + T_{\alpha\beta}^{\mathcal{M}} = 0 \quad \implies \quad T_{++} + T_{++}^{\mathcal{M}} = 0 \quad \text{and} \quad T_{--} + T_{--}^{\mathcal{M}} = 0 \quad (22)$$

2.5 Classical Geodesic Solutions in AdS₃

Here, we look for classical solutions that depend only on $\tau = x^+ + x^-$, where the string has no spatial extension and reduces to a point particle moving along the AdS₃ geodesic.

For timelike geodesics, consider an ansatz solution,

$$g_+(x^+) = Ue^{iv_+(x^+)\sigma_2}, \quad g_-(x^-) = e^{iu_-(x^-)\sigma_2}V \quad \text{where} \quad U, V \in \text{SL}(2, \mathbb{R}) \quad (23)$$

$$\implies J_R(x^+) = -k(\partial_+v_+)(UT^3U^{-1}) \quad \text{and} \quad J_L(x^-) = -k(\partial_-u_-)V^{-1}T^3V \quad (24)$$

Then, the chiral components of the energy-momentum tensors from the AdS₃ sector are,

$$T_{++}(x^+) = -k(\partial_+v_+)^2 \quad \text{and} \quad T_{--}(x^-) = -k(\partial_-u_-)^2 \quad (25)$$

If the internal $\mathcal{M}$ CFT has conformal weights h , then, the Virasoro constraints give,

$$-k(\partial_+v_+)^2 + h = 0 \quad \text{and} \quad -k(\partial_-u_-)^2 + h = 0 \implies (\partial_+v_+) = \pm\sqrt{\frac{h}{k}} = (\partial_-u_-) \quad (26)$$

$$\implies v_+ = \frac{\alpha}{2}x^+ \quad \text{and} \quad u_- = \frac{\alpha}{2}x^- \quad \text{where} \quad \alpha = \pm\sqrt{\frac{4h}{k}} \quad (27)$$

$$\implies g(x^+, x^-) = Ue^{i\alpha\tau\sigma_2}V = U \begin{pmatrix} \cos(\alpha\tau) & \sin(\alpha\tau) \\ -\sin(\alpha\tau) & \cos(\alpha\tau) \end{pmatrix} V \quad (28)$$

This is a point particle-like solution evolving along a trajectory in AdS₃ parameterized by τ . In the special case $U = V = 1$, the solution corresponds to a particle located at the center of AdS₃ where, in terms of global coordinates, it is $t = \alpha\tau$ and $\rho = 0$. The metric along the geodesic is $ds^2 = -\alpha^2 d\tau^2 < 0$. So the trajectory is time-like. Then, a general solution of the form $g = Ue^{i\alpha\tau\sigma_2}V$ is just an isometry transformation. Thus, even then, it describes a time-like geodesic in AdS₃. The Monodromy matrix for this solution is

$$M = e^{i\alpha\pi\sigma_2} = \begin{pmatrix} \cos(\alpha\pi) & \sin(\alpha\pi) \\ -\sin(\alpha\pi) & \cos(\alpha\pi) \end{pmatrix} \implies \text{Tr}(M) = 2\cos(\alpha\pi) \implies |\text{Tr}(M)| < 2 \quad (29)$$

Hence, this solution corresponds to the Elliptic conjugacy class of $\text{SL}(2, \mathbb{R})$ which describes short strings oscillating near the center of AdS₃.

For spacelike geodesics, that correspond to the Hyperbolic conjugacy class of $\text{SL}(2, \mathbb{R})$,

$$g_+(x^+) = Ue^{v_+(x^+)\sigma_3}, \quad g_-(x^-) = e^{u_-(x^-)\sigma_3}V \quad \text{where} \quad U, V \in \text{SL}(2, \mathbb{R}) \quad (30)$$

$$\implies J_R(x^+) = -k(\partial_+v_+)(UT^1U^{-1}) \quad \text{and} \quad J_L(x^-) = -k(\partial_-u_-)V^{-1}T^1V \quad (31)$$

Then, the chiral components of the energy-momentum tensors from the AdS₃ sector are,

$$T_{++}(x^+) = k(\partial_+v_+)^2 \quad \text{and} \quad T_{--}(x^-) = k(\partial_-u_-)^2 \quad (32)$$

Then, the Virasoro constraints give, $v_+ = \frac{\alpha}{2}x^+$ and $u_- = \frac{\alpha}{2}x^-$ where $\alpha = \pm\sqrt{-\frac{4h}{k}}$.

$$\implies g(x^+, x^-) = Ue^{\alpha\tau\sigma_3}V \quad (33)$$

In the case $U = V = 1$, the solution corresponds to a space-like geodesic of AdS_3 given by $\rho = \alpha\tau$ and $t = \phi = 0$. The trajectory is therefore a straight radial line emanating from the center of AdS_3 . The metric along the geodesic is $ds^2 = d\rho^2 > 0$ and is space-like. The Monodromy matrix for this solution is

$$M = e^{\alpha\pi\sigma_3} = \begin{pmatrix} e^{\alpha\pi} & 0 \\ 0 & e^{-\alpha\pi} \end{pmatrix} \implies \text{Tr}(M) = 2 \cosh(\alpha\pi) \implies |\text{Tr}(M)| > 2 \quad (34)$$

Hence, this solution describes long strings reaching toward the spatial boundary of AdS_3 .

For null geodesics, whose solutions correspond to the parabolic conjugacy class of $\text{SL}(2, \mathbb{R})$ when we have the ansatz to have $T^\pm$ as the nilpotent raising / lowering generators whose $(T^+)^2 = (T^-)^2 = 0$. This type of solution is given by

$$g^{(\pm)}(x^+, x^-) = U e^{\alpha\tau\sigma^\pm} V, \quad U, V \in \text{SL}(2, \mathbb{R}), \quad \alpha = \text{some constant} \quad (35)$$

These solutions have $T_{\pm\pm} = h = 0 = ds^2$. So, this solution represents a string moving along the null direction in AdS_3 . The Monodromy matrix is given by,

$$M = e^{2\alpha\pi T^\pm} = 1 + 2\alpha\pi T^\pm = \begin{pmatrix} 1 + \alpha\pi & \pm\alpha\pi \\ \mp\alpha\pi & 1 - \alpha\pi \end{pmatrix} \implies |\text{Tr}(M)| = 2 \quad (36)$$

2.6 Spectral flow of Classical Geodesic Solutions

A distinctive feature of the $\text{SL}(2, \mathbb{R})$ WZW model is the existence of a family of transformations known as *spectral flow* which acts on a given solution to produce another solution classically. We choose a particular form of spectral flow generated by $T^3 = -i\sigma_2/2$.

$$\begin{aligned} \tilde{g}_+(x^+) &\longrightarrow g_+(x^+) = e^{\frac{i}{2}w_R x^+ \sigma_2} \tilde{g}_+(x^+) \quad , \quad \tilde{g}_-(x^-) \longrightarrow g_-(x^-) = \tilde{g}_-(x^-) e^{\frac{i}{2}w_L x^- \sigma_2} \\ \implies g(x^+, x^-) &= g_+(x^+) g_-(x^-) = e^{\frac{i}{2}w_R x^+ \sigma_2} \tilde{g}(x^+, x^-) e^{\frac{i}{2}w_L x^- \sigma_2} \quad , \quad w_R, w_L \in \mathbb{Z} \end{aligned} \quad (37)$$

Under spectral flow, the ρ coordinate is unchanged, but the other coordinates change as,

$$\begin{aligned} t &= \tilde{t} + \frac{1}{2}(w_R x^+ + w_L x^-) = \tilde{t} + \frac{1}{2}\left((w_R + w_L)\tau + (w_R - w_L)\sigma\right) \\ \phi &= \tilde{\phi} + \frac{1}{2}(w_R x^+ - w_L x^-) = \tilde{\phi} + \frac{1}{2}\left((w_R - w_L)\tau + (w_R + w_L)\sigma\right) \end{aligned} \quad (38)$$

For the diagonal spectral flow $w_L = w_R = w$, we get, $t = \tilde{t} + w\tau$, $\rho = \tilde{\rho}$ and $\phi = \tilde{\phi} + w\sigma$. So, the transformed solution acquires winding number w around the angular direction of AdS_3 . If the original solution has a monodromy matrix $\tilde{M}$, then, under spectral flow,

$$M \rightarrow e^{iw\pi\sigma_2} M = (-1)^w M \quad (39)$$

Spectral flow modifies the monodromy of a classical solution by a central element $\pm \mathbf{I}$. But, $|\text{Tr}(\tilde{M})| = |\text{Tr}(M)|$. Hence, the conjugacy class (elliptic/hyperbolic/parabolic) and local geometry are unchanged.

Under spectral flow, the currents and the energy-momentum tensor transform as,

$$J_R^3(x^+) = \tilde{J}_R^3(x^+) + \frac{k}{2}w_R \quad , \quad J_R^\pm(x^+) = e^{\mp i w_R x^+} \tilde{J}_R^\pm(x^+) \quad (\text{similar for } J_L)$$

$$T_{++} = \tilde{T}_{++} - w_R \tilde{J}_R^3 - \frac{k}{4}w_R^2 \quad \text{and} \quad T_{--} = \tilde{T}_{--} - w_L \tilde{J}_L^3 - \frac{k}{4}w_L^2$$
(40)

Expanding the currents and energy-momentum tensors in Fourier modes,

$$J_R^a(x^+) = \sum_{n \in \mathbb{Z}} J_n^a e^{-inx^+} \quad \text{and} \quad J_L^a(x^-) = \sum_{n \in \mathbb{Z}} \bar{J}_n^a e^{-inx^-}$$

$$T_{++}(x^+) = \sum_{n \in \mathbb{Z}} L_n e^{-inx^+} \quad \text{and} \quad T_{--}(x^-) = \sum_{n \in \mathbb{Z}} \bar{L}_n e^{-inx^-}$$
(41)

Spectral flow acts on the modes as,

$$J_n^3 = \tilde{J}_n^3 + \frac{k}{2}w_R \delta_{n,0} \quad , \quad J_n^\pm = \tilde{J}_{n \mp w_R}^\pm \quad , \quad L_n = \tilde{L}_n - w_R \tilde{J}_n^3 - \frac{k}{4}w_R^2 \delta_{n,0}$$

$$\bar{J}_n^3 = \tilde{\bar{J}}_n^3 + \frac{k}{2}w_L \delta_{n,0} \quad , \quad \bar{J}_n^\pm = \tilde{\bar{J}}_{n \mp w_L}^\pm \quad , \quad \bar{L}_n = \tilde{\bar{L}}_n - w_L \tilde{\bar{J}}_n^3 - \frac{k}{4}w_L^2 \delta_{n,0}$$
(42)

Short strings are the spectral flow of timelike geodesics, $\tilde{g}(\tau) = U e^{i\alpha\tau\sigma_2} V$, $\alpha = \pm\sqrt{\frac{4h}{k}}$, $\tilde{T}_{\pm\pm} = -\frac{k}{4}\alpha^2$. To find the space-time energy after spectral flow, we shall impose the Virasoro constraint for the spectrally flowed energy-momentum tensor, which gives,

$$J_0^3 = \frac{1}{w_R} \left(h - \frac{k}{4}\alpha^2 \right) + \frac{k}{4}w_R \quad \text{and} \quad \bar{J}_0^3 = \frac{1}{w_L} \left(h - \frac{k}{4}\alpha^2 \right) + \frac{k}{4}w_L$$
(43)

Now, for $U = V^{-1} = e^{\frac{1}{2}\rho_0\sigma_3}$, and diagonal spectral flow $w_L = w_R = w$ we get,

$$\tan(t) = \frac{\tan(w\tau) + \tan(\alpha\tau)/\cosh(\rho_0)}{1 - \tan(w\tau)\tan(\alpha\tau)/\cosh(\rho_0)} \quad \text{and} \quad \sinh(\rho) e^{i\phi} = i \sinh(\rho_0) \sin(\alpha\tau) e^{iw\sigma}$$
(44)

$\implies 0 \leq \sinh(\rho) \leq \sinh(\rho_0)$. So, ρ is bounded for all τ . The explicit J^3 currents are,

$$J_0^3 = \bar{J}_0^3 = \frac{k}{2} \left(\alpha \cosh(\rho_0) + w \right) \implies \alpha = -w \cosh(\rho_0) + \sqrt{w^2 \sinh^2(\rho_0) + \frac{4h}{k}}$$

$$\implies E_s = J_0^3 + \bar{J}_0^3 = k \cosh(\rho_0) \sqrt{w^2 \sinh^2(\rho_0) + \frac{4h}{k}} - kw \sinh^2(\rho_0)$$
(45)

$$E_s(\rho_0 = 0) = 2\sqrt{hk} \quad \text{and} \quad E_s(\rho_0 \rightarrow \infty) = \frac{kw}{2} + \frac{2h}{w}$$

So, this solution is a bound state of the string in AdS_3 . Then, at the critical value of h ,

$$h_{\text{crit}} = \frac{k w^2}{4} \quad \text{we get} \quad t = w\tau \quad , \quad \phi = w\sigma \quad , \quad \rho = \rho_0 \quad (\text{fixed radius}) \quad \text{and} \quad E_s(h_{\text{crit}}) = kw$$
(46)

Long strings are the spectral flow of space-like geodesics, $\tilde{g}(\tau) = U e^{\alpha\tau\sigma_3} V$, $\alpha = \pm\sqrt{-\frac{4h}{k}}$, $T_{\pm\pm} = \frac{k\alpha^2}{4}$. Following the same procedure, we get,

$$J_0^3 = \frac{1}{w_R} \left(h + \frac{k}{4}\alpha^2 \right) + \frac{k}{4}w_R \quad \text{and} \quad \bar{J}_0^3 = \frac{1}{w_L} \left(h + \frac{k}{4}\alpha^2 \right) + \frac{k}{4}w_L$$
(47)

When, $U = V^{-1} = e^{-\frac{1}{2}\rho_0\sigma_1}$, the solution in terms of global coordinates is,

$$\tan(t) = \frac{\tan(w\tau) + \sinh(\rho_0) \tanh(\alpha\tau)}{1 - \sinh(\rho_0) \tanh(\alpha\tau) \tan(w\tau)} \quad \text{and} \quad \sinh(\rho) e^{i\phi} = e^{iw\sigma} \cosh(\rho_0) \sinh(\alpha\tau)$$
(48)

Thus, at the infinite past $t \rightarrow -\infty$, we see that the string is circular and has infinite radius located at the spatial boundary of AdS_3 . Then the string collapses to a point at $t = \tau = 0$, and then expands again toward the boundary as $t \rightarrow \infty$. This is the long-string solution. The explicit J^3 currents after diagonal spectral flow is, $J_R^3 = J_L^3 = \frac{k}{2}(\alpha \sinh(\rho_0) + w)$, which gives $\alpha = w \sinh(\rho_0) - \sqrt{w^2 \cosh^2(\rho_0) - \frac{4h}{k}}$. This results in,

$$E_l = kw \cosh^2(\rho_0) - k \sinh(\rho_0) \sqrt{w^2 \cosh^2(\rho_0) - \frac{4h}{k}} \quad (49)$$

$$E_l(\rho_0 = 0) = kw \quad \text{and} \quad E_l(\rho_0 \rightarrow \infty) = \infty \quad (E_l \text{ is unbounded above})$$

At $h = h_{\text{crit}} = \frac{kw^2}{4}$, the energy of the long-string solution becomes $E_l(h_{\text{crit}}) = kw$ which coincides with $E_s(h_{\text{crit}})$. Moreover, at h_{crit} , $\partial E_s / \partial h = \partial E_l / \partial h = 2/w$. Thus, the two branches meet smoothly in the (h, E) plane. So, there can be a smooth transition between short- and long-string solutions at $h = h_{\text{crit}}$ without any discontinuity in its energy.

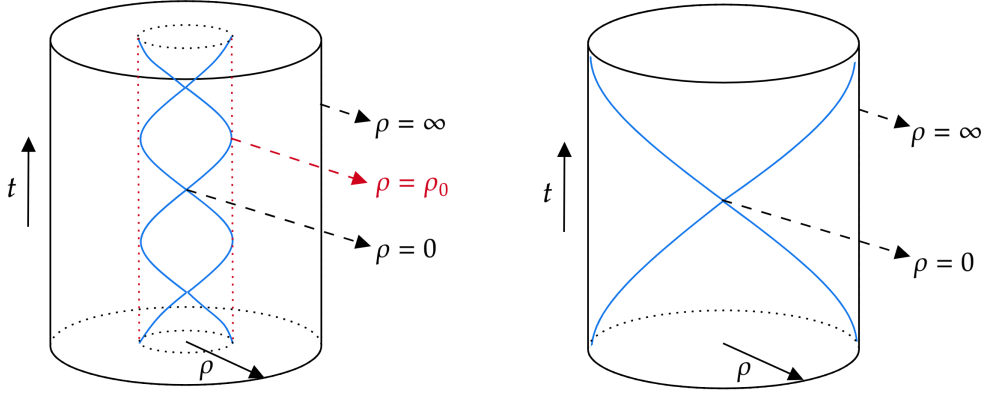


Figure 1: (left) Spectral flow of timelike geodesic giving short string (bounded) solution. (right) Spectral flow of space-like geodesics giving long string (unbounded) solution.

2.7 Semiclassical Analysis Near the Center of AdS_3

To understand the origin of spectral flow from a semiclassical perspective, we consider string dynamics near the center of AdS_3 , where $\rho \ll 1$. Since the angular coordinate ϕ becomes ill-defined at $\rho = 0$, we introduce $X^1 = \rho \cos(\phi)$ and $X^2 = \rho \sin(\phi)$ and the complex field $\Phi = X^1 + iX^2 = \rho e^{i\phi}$. Expanding the AdS_3 metric and the NS-NS B -field to quadratic order in ρ by keeping only quadratic terms in the induced worldsheet metric, and in the static gauge $t = \tau$, the Nambu-Goto action reduces to,

$$S_{\text{NG}} = \frac{\sqrt{4hk}}{2\pi} \int d\tau d\sigma \left[1 - \frac{1}{2} |\partial_\tau \Phi|^2 + \frac{1}{2A^2} |\partial_\sigma \Phi - iA\Phi|^2 \right], \quad A = \sqrt{\frac{4h}{k}} \quad (50)$$

The spatial derivative appears through the covariant derivative $D_\sigma = \partial_\sigma - iA$, which is like the minimal coupling of a complex scalar of unit charge to a gauge field. The ‘charge’ originates from the NS-NS B -field of AdS_3 .

$$\text{Eqn. of Motion : } \partial_{\hat{\tau}}^2 \Phi - (\partial_\sigma - iA)^2 \Phi = 0 \quad , \quad \hat{\tau} = \frac{\tau}{A} \quad (51)$$

$$\text{Solution : } \Phi(\hat{\tau}, \sigma) = e^{iA\sigma} (f(\hat{\tau} + \sigma) + g(\hat{\tau} - \sigma))$$

The periodic boundary condition of $\Phi(\hat{\tau}, \sigma + 2\pi) = \Phi(\hat{\tau}, \sigma)$, results in,

$$f(\hat{x}^+ + 2\pi) = e^{-2i\pi A} f(\hat{x}^+) \quad \text{and} \quad g(\hat{x}^- - 2\pi) = e^{-2i\pi A} g(\hat{x}^-) \quad (52)$$

For this kind of twisted periodicity, these functions can be Fourier expanded as,

$$f(\hat{x}^+) = \sum_{n \in \mathbb{Z}} \frac{a_n^\dagger}{n - A} e^{i(n-A)\hat{x}^+} \quad \text{and} \quad g(\hat{x}^-) = \sum_{n \in \mathbb{Z}} \frac{b_n}{n - A} e^{-i(n-A)\hat{x}^-} \quad (53)$$

Canonical quantization then yields oscillator commutators proportional to $(n - A)$ as,

$$[a_n, a_m^\dagger] = (n - A)\delta_{nm} \quad \text{and} \quad [b_n, b_m^\dagger] = (n - A)\delta_{nm} \quad (54)$$

Consequently, the definition of creation and annihilation operators changes whenever A crosses an integer value $w = \lfloor A \rfloor$, which measures the resulting shift of the oscillator vacuum (Fermi/Bose sea level) and labels the corresponding spectrally flowed sector. We can also show that $J_n^+ \sim -ik a_n$, $J_n^- \sim ik a_{-n}^\dagger$, $\bar{J}_n^+ = -ik b_n$, and $\bar{J}_n^- = ik b_{-n}^\dagger$. Thus,

$$J_n^+|0\rangle = \bar{J}_n^+|0\rangle = 0 \quad (\text{when } n > A) \quad \text{and} \quad J_{-n}^-|0\rangle = \bar{J}_{-n}^-|0\rangle = 0 \quad (\text{when } n < A) \quad (55)$$

This is not the standard highest-weight condition of an affine algebra, since the annihilation threshold is shifted from the usual $n > 0$. So, to restore the highest weight state form for the vacuum, let us introduce a new set of current modes $\tilde{J}_n^\pm = J_{n \pm w}^\pm$. Then,

$$\tilde{J}_n^+|0\rangle = J_{n+w}^+|0\rangle = 0 \quad (\text{when } n > 0) \quad \text{and} \quad \tilde{J}_n^-|0\rangle = J_{n-w}^-|0\rangle = 0 \quad (\text{when } n \geq 0) \quad (56)$$

Thus, the vacuum is the highest-weight state with respect to the shifted currents $\tilde{J}_n^\pm$. So, $w \in \mathbb{Z}$ labels in-equivalent spectral-flow sectors of the affine $\widehat{\mathfrak{sl}}(2, \mathbb{R})$ algebra. In this way, spectral flow emerges semiclassically as a reorganization of the oscillator spectrum, corresponding to a shift of the vacuum relative to the unflowed highest-weight representation.

3 Quantum theory of Strings in AdS₃

3.1 Spectrum generating algebra of $\text{SL}(2, \mathbb{R})$ WZW model

Equations of motion of WZW model imply the existence of Lie-algebra-valued conserved currents $J_{R,L}$. Expanding them in $\{T^a\}$ basis as $J_{R,L} = \kappa_{ab} J_{R,L}^a T^b$, where the basis satisfy $[T^a, T^b] = f^{ab}_c T^c$, with f^{ab}_c as structure constants of the Lie-algebra, one can derive their equal-time Poisson brackets from the canonical symplectic structure of the theory. Upon Fourier expansion and canonical quantization, the current modes generate an affine Kac–Moody algebra of the form, $[J_m^a, J_n^b] = f^{ab}_c J_{m+n}^c + \frac{k}{2} m \kappa^{ab} \delta_{m+n,0}$. Then, for the $\text{SL}(2, \mathbb{R})$ WZW model, we get the corresponding affine $\widehat{\mathfrak{sl}}(2, \mathbb{R})_k \times \widehat{\mathfrak{sl}}(2, \mathbb{R})_k$ algebra,

$$\begin{aligned} [J_m^3, J_n^3] &= -\frac{k}{2} m \delta_{m+n,0} \quad , \quad [J_m^3, J_n^\pm] = \pm J_{n+m}^\pm \quad \text{and} \quad [J_m^+, J_n^-] = k m \delta_{m+n,0} - 2J_{n+m}^3 \\ [\bar{J}_m^3, \bar{J}_n^3] &= -\frac{k}{2} m \delta_{m+n,0} \quad , \quad [\bar{J}_m^3, \bar{J}_n^\pm] = \pm \bar{J}_{n+m}^\pm \quad \text{and} \quad [\bar{J}_m^+, \bar{J}_n^-] = k m \delta_{m+n,0} - 2\bar{J}_{n+m}^3 \end{aligned} \quad (57)$$

The Sugawara energy-momentum tensor is identical to the classical stress tensor, but with a quantum normalization such that the currents have conformal weight 1. Accordingly,

$$T_{++}(x^+) = \frac{1}{k-2} \kappa_{ab} :J_R^a(x^+) J_R^b(x^+): \quad \text{and} \quad T_{--}(x^+) = \frac{1}{k-2} \kappa_{ab} :J_L^a(x^+) J_L^b(x^+): \quad (58)$$

Substituting the mode expansion of the currents in the above expression and comparing with the $T_{\pm\pm}$ mode expansion, the Virasoro generators are obtained as

$$L_n = \frac{1}{k-2} \sum_{m \in \mathbb{Z}} \kappa_{ab} : J_{n-m}^a J_m^b : \quad \text{and} \quad \bar{L}_n = \frac{1}{k-2} \sum_{m \in \mathbb{Z}} \kappa_{ab} : \bar{J}_{n-m}^a \bar{J}_m^b : \quad (59)$$

Using the $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ algebra, one finds, $[L_m, J_n^a] = -n J_{n+m}^a$ (same for $\bar{L}_n, \bar{J}_m^a$) (60) which shows that the currents transform as conformal fields of weight one. And also,

$$[L_m, L_n] = (m-n) L_{m+n} + \frac{c}{12} m(m^2-1) \delta_{m+n,0} \quad , \quad c = \frac{3k}{k-2} \quad (\text{same for } \bar{L}_n) \quad (61)$$

The spectral flow transformation shifts the current and Virasoro modes without altering the structure of their commutation relations or the central charge. The spectrum-generating algebra is therefore invariant under spectral flow, demonstrating that it is an automorphism of the theory, which maps one representation of the theory to another and plays a crucial role in generating the complete physical spectrum of strings on AdS_3 .

3.2 Representation theory of the affine $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ current algebra

The complete Hilbert space is obtained by the representations of the $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k \times \widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ algebra. In string theory on AdS_3 , the full spectrum decomposes into sectors labelled by the spectral-flow parameter $w \in \mathbb{Z}$, which is interpreted as the string winding number. From now on, we will discuss only the right sector, as both chiral sectors are analogous.

3.2.1 Unitary Representations of the global $\mathfrak{sl}(2, \mathbb{R})$ algebra

The full Hilbert space can be organized by the representations of the $\mathfrak{sl}(2, \mathbb{R})$ sub-algebra of the zero modes of the affine currents that satisfy $[J_0^3, J_0^\pm] = \pm J_0^\pm$ and $[J_0^+, J_0^-] = -2J_0^3$. The zero-mode representations are characterized by the quadratic Casimir operator,

$$C_2 = \kappa_{ab} J_0^a J_0^b = \frac{1}{2} (J_0^+ J_0^- + J_0^- J_0^+) - (J_0^3)^2 \quad (62)$$

C_2 commutes with all current modes J_n^a and Virasoro generators L_n , while L_n commute with all zero modes J_0^a . So, $\{C_2, J_0^3, L_0\}$ forms a convenient commuting set of operators. Here L_0 measures the conformal weight of a state, while C_2 labels the underlying global $\mathfrak{sl}(2, \mathbb{R})$ representation. Restricting to positive-energy (unflowed) representations, whose L_0 spectrum is bounded from below, there exist lowest-weight states called as affine primaries, denoted by $|j, m\rangle$ which are annihilated by all positive current modes, $J_{n>0}^a |j, m\rangle = 0$ and are simultaneous eigen-kets of C_2, J_0^3, L_0 as,

$$C_2 |j, m\rangle = -j(j-1) |j, m\rangle \quad , \quad J_0^3 |j, m\rangle = m |j, m\rangle \quad , \quad L_0 |j, m\rangle = -\frac{j(j-1)}{k-2} |j, m\rangle \quad (63)$$

The last relation follows from, $L_0 = \frac{C_2}{k-2} + \frac{1}{k-2} \sum_{n=1}^{\infty} (J_{-n}^+ J_n^- + J_{-n}^- J_n^+ - 2J_{-n}^3 J_n^3)$ (64)

Hence, the conformal weight of an affine primary is completely determined by C_2 . Using $\mathfrak{sl}(2, \mathbb{R})$ algebra and the Casimir eigenvalue, one finds the action of $J_0^\pm$ on $|j, m\rangle$ as,

$$J_0^\pm |j, m\rangle = \sqrt{(m \mp j \pm 1)(m \pm j)} |j, m \pm 1\rangle \quad (65)$$

Thus, repeated action of $J_0^\pm$ generates the entire $\text{SL}(2, \mathbb{R})$ representation labelled by j .

The unitary irreducible representations of the global $\mathfrak{sl}(2, \mathbb{R})$ algebra are obtained by requiring positivity of norms $\|J_0^\pm |j, m\rangle\|^2 \geq 0$, together with the ladder-operator action. This yields the following classes of unitary representations:

1. **Discrete series representation** $\mathcal{D}_j^\pm$ ('+' $\rightarrow$ Lowest-weight, '-' $\rightarrow$ Highest-weight):

$$\mathcal{D}_j^\pm = \{|j, m\rangle : m = \pm j, \pm(j+1), \pm(j+2), \dots\} \quad , \quad j > 0$$

2. **Principal continuous series** $\mathcal{C}_j^\alpha$:

$$\mathcal{C}_j^\alpha = \{|j, \alpha, m\rangle : m = \alpha + n, n \in \mathbb{Z}, j = 1/2 + is, s \in \mathbb{R}, \alpha \in [0, 1)\}$$

3. **Complementary series** $\mathcal{E}_j^\alpha$:

$$\mathcal{E}_j^\alpha = \{|j, \alpha, m\rangle : m = \alpha + n, n \in \mathbb{Z}, 1/2 < j < 1, |\alpha - 1/2| \geq j - 1/2\}$$

4. **Trivial representation** $\mathcal{I}$: $\mathcal{I} = \{|j = 0, m = 0\rangle\}$

Harmonic analysis of Quantum Mechanics of a particle in AdS_3 : To identify the representations that appear in the physical spectrum of the WZW model, consider the large- k limit, where the curvature of AdS_3 becomes small. The theory reduces to the quantum mechanics of a particle propagating on AdS_3 . The corresponding Hilbert space $\mathcal{H} = L^2(\text{AdS}_3)$, is the space of square-integrable functions on AdS_3 , which can be decomposed into unitary representations of the isometry group $\text{SL}(2, \mathbb{R})_L \times \text{SL}(2, \mathbb{R})_R$. A convenient basis for $\mathcal{H}$ is provided by the matrix elements $F_{m, \bar{m}}^j(g) = \langle j, m | g | j, \bar{m} \rangle$, where $g \in \text{SL}(2, \mathbb{R})$. The Laplacian on AdS_3 is related to the quadratic Casimir by $\nabla^2 = -C_2$ with $C_2 = -j(j-1)$, the representation label j determines the asymptotic behavior of the radial wavefunctions $R^{(j)}(\rho)$. For the discrete series $\mathcal{D}_j^\pm$, one finds $R^{(j)}(\rho) \sim e^{-2j\rho}$, which is square-integrable only for $j > \frac{1}{2}$, corresponding to normalizable bound states localized in the bulk of AdS_3 . Then, for the continuous series, the asymptotic behavior $R^{(j=1/2+is)}(\rho) \sim e^{-\rho} e^{\pm is\rho}$ describes oscillatory plane-wave states, which are delta-function normalizable scattering states. So, the harmonic analysis of $L^2(\text{AdS}_3)$ gives,

$$\mathcal{L}^2(\text{AdS}_3) = \bigoplus_{j > \frac{1}{2}} \mathcal{D}_j^\pm \otimes \mathcal{D}_j^\pm \oplus \int ds \mathcal{C}_{\frac{1}{2}+is}^\alpha \otimes \mathcal{C}_{\frac{1}{2}+is}^\alpha \quad (66)$$

The complementary series $\mathcal{E}_j^\alpha$ does not appear in the standard Hilbert space, as its states are not normalizable with respect to the usual inner product. Their inclusion would require a j -dependent definition of the norm, which is a non-standard quantization.

3.2.2 Representations of the current algebra and no-ghost theorem

Representations of the affine algebra $\widehat{\mathfrak{sl}}(2, \mathbb{R})_k$ are obtained by promoting states in the Hilbert space $\mathcal{H}$, built from the unitary representations $\mathcal{D}_j^\pm$ and $\mathcal{C}_j^\alpha$ of the global $\mathfrak{sl}(2, \mathbb{R})$ algebra, to affine primary states and generating descendants through the action of negative current modes. The resulting affine modules are denoted by $\hat{\mathcal{D}}_j^\pm$ and $\hat{\mathcal{C}}_j^\alpha$, and the corresponding representation space takes the form

$$\hat{\mathcal{H}} = \text{Span} \{ J_{-n_1}^{a_1} J_{-n_2}^{a_2} \dots |\text{Primary}\rangle \} \quad , \quad |\text{Primary}\rangle \in \mathcal{H} \quad (67)$$

Also, since $[C_2, J_n^a] = 0$, every descendant state carries the same Casimir eigenvalue as the affine primary and inherits the same representation label j .

Affine $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ representations generally contain negative-norm states which are not physical. So, the physical states $|\psi\rangle$ are the ones that obey the Virasoro constraints,

$$(L_n + \mathcal{L}_n^{\mathcal{M}} - \delta_{n,0})|\psi\rangle = 0 \quad , \quad n \geq 0 \quad (68)$$

The J^3 current modes form a $\widehat{\mathfrak{u}(1)}_k$ affine algebra of a time-like bosonic field whose Virasoro modes are $L_n^{(J^3)} = -\frac{1}{k} \sum_{m \in \mathbb{Z}} : J_{n-m}^3 J_m^3 :$. Decomposing $L_n = L_n^{(J^3)} + \widehat{L}_n$, one finds that every physical state which satisfy the above Virasoro constraints can be represented, modulo null states, by those states that satisfy $J_{n>0}^3 |\psi\rangle = 0$ and hence belonging to the coset $\text{SL}(2, \mathbb{R})/U(1)$ without any time-like excitations. For this to be true, we require $k > 2$ and $c_{\text{total}} = 26$, and the physical states $|\psi\rangle$ have to be taken from the descendants of discrete series $\mathcal{D}_j^\pm$ with $\frac{1}{2} \leq j \leq \frac{k}{2}$ or from the continuous series $C_{j=\frac{1}{2}+is}^\alpha$, $s \in \mathbb{R}$. This is the no-ghost theorem for the unflowed sector.

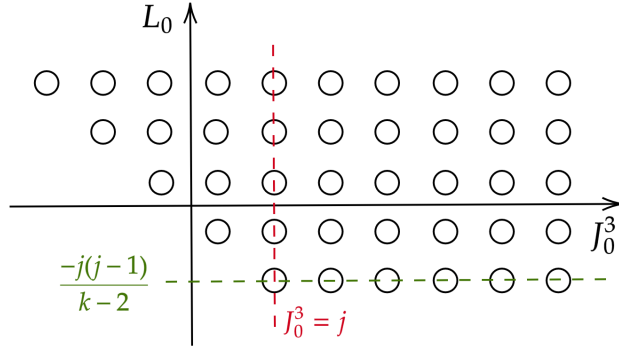


Figure 2: Weight diagram in unflowed lowest weight representation $\widehat{\mathcal{D}}_j^+$ built on the primary state $|j, m = j\rangle$. States are organized according to (J_0^3, L_0) values. The zero mode J_0^+ generates the global $\text{SL}(2, \mathbb{R})$ descendants with fixed $L_0^{\min}$. Affine descendants are obtained by acting with negative current modes J_{-n}^a , which raise the L_0 value by n and shift the J_0^3 value according to $J_{-n}^+ : m \rightarrow m + 1$, $J_{-n}^- : m \rightarrow m - 1$, $J_{-n}^3 : m \rightarrow m$. Thus, this is a positive-energy representation with L_0 bounded below.

Unlike unflowed representations, the spectrally flowed modules need not have L_0 bounded below which are denoted by $\widehat{\mathcal{D}}_{\tilde{j}}^{\pm, w}$, $\widehat{\mathcal{C}}_{\tilde{j}}^{\alpha, w}$ where $\tilde{j}$ is the spin before the spectral flow.

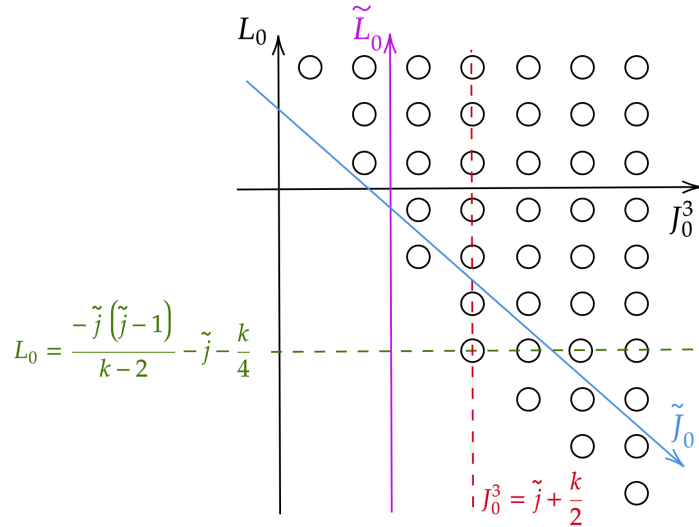


Figure 3: Spectrally flowed $\widehat{\mathcal{D}}_j^{+, w=1}$ representation.

The above weight-diagram illustrates the spectrally flowed discrete representation $\hat{\mathcal{D}}_j^{+,w=1}$. Under spectral flow, the current and Virasoro zero modes are redefined such that L_0 gets mixed with the J_0^3 value. As a result, states that lay on horizontal lines of fixed $\tilde{L}_0$ in the unflowed representation appear as diagonal lines of slope -1 in the (J_0^3, L_0) plane. The flowed spectrum is no longer bounded below since L_0 decreases as J_0^3 increases. Nevertheless, the J_0^3 spectrum and hence the space-time energy remain bounded below in a fixed L_0 level which is selected by the Virasoro constraint $L_0 + h = 1$.

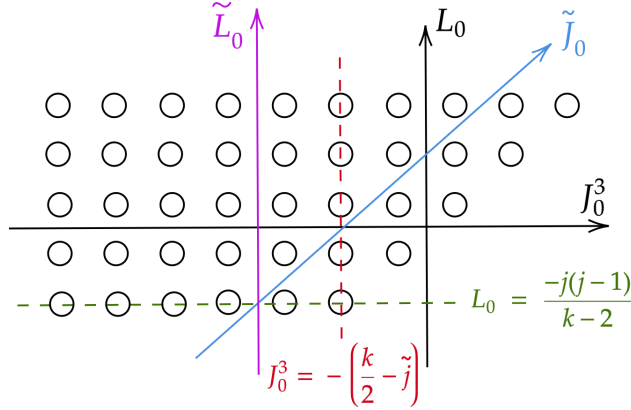


Figure 4: Spectral flow with $w = -1$ on $\hat{\mathcal{D}}_j^+$ results in $\hat{\mathcal{D}}_{\tilde{j}}^{+,w=-1} = \hat{\mathcal{D}}_{j=\frac{k}{2}-\tilde{j}}^-$. The lowest-weight state of $\hat{\mathcal{D}}_j^+$ is mapped to a highest-weight state of $\hat{\mathcal{D}}_{j=\frac{k}{2}-\tilde{j}}^-$. Thus, $L_0^{\min} = \tilde{L}_0^{\min}$. So, even the spectrally-flowed representation remains bounded below in L_0 .

Similarly, spectral flow with $w = +1$ maps $\hat{\mathcal{D}}_j^-$ to $\hat{\mathcal{D}}_{\tilde{j}}^{-,w=+1} = \hat{\mathcal{D}}_{j=\frac{k}{2}-\tilde{j}}^+$, whose L_0 spectrum is bounded below. Generally,

$$\hat{\mathcal{D}}_{\tilde{j}}^{\pm,w} = \hat{\mathcal{D}}_{j=\frac{k}{2}-\tilde{j}}^{\mp,w\pm 1} \quad (69)$$

Since spectral flow is a symmetry of the theory, normalizability and the no-ghost condition must be preserved under this identification. Requiring both $\tilde{j}$ and $j = \frac{k}{2} - \tilde{j}$ to satisfy the unitarity bound therefore yields the truncation

$$\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2} \quad \text{and} \quad \frac{1}{2} \leq j \leq \frac{k-1}{2} \quad (70)$$

So, $\hat{\mathcal{D}}_{\tilde{j}}^{+,w}$ with $w \geq 1$ or $w \leq -2$ and $\hat{\mathcal{D}}_{\tilde{j}}^{-,w}$ with $w \geq 2$ or $w \leq -1$ becomes unbounded below in L_0 . Likewise, for the principal continuous representations $\hat{\mathcal{C}}_{j=\frac{1}{2}+is}^\alpha$, any nontrivial spectral flow ($w \neq 0$) yields a representation whose L_0 spectrum is unbounded from below.

The no-ghost theorem extends naturally to the spectrally flowed representations $\hat{\mathcal{D}}_{\tilde{j}}^{\pm,w}$ and $\hat{\mathcal{C}}_{\tilde{j}}^{\alpha,w}$. Since spectral flow is an automorphism of the spectrum-generating algebra, it preserves the structure of the Hilbert space. Imposing the physical-state conditions with the flowed Virasoro generators, together with the flowed mass-shell condition, one finds that every physical state is equivalent, modulo null states, to a state satisfying $J_{n>0}^3|\psi\rangle = 0$, where J_n^3 denotes the flowed current. Consequently, negative-norm states decouple exactly as in the unflowed theory, and the spectrally flowed sectors remain ghost-free under the same conditions as their unflowed counterparts.

3.3 Physical Spectra of Strings in AdS₃

Physical states are built from representations of $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ tensored with an internal CFT of conformal weight h . All flowed sectors must be included to get the full spectrum. For an unflowed state, $|\tilde{j}, \tilde{m}, \tilde{N}, h\rangle$ where $\tilde{N}$ is the oscillator level, the new physical-state condition in the w -th flowed sector becomes,

$$\begin{aligned} -\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - w\tilde{m} - \frac{k}{4}w^2 = 1 &\implies J_0^3 = \tilde{m} + \frac{k}{2}w \\ \implies J_0^3 = \frac{k}{4}w + \frac{1}{w} \left(-\frac{\tilde{j}(\tilde{j}-1)}{k-2} + \tilde{N} + h - 1 \right) \end{aligned} \quad (71)$$

For discrete representations, $\tilde{m} = \tilde{j} + q$, where $q \geq -\tilde{N}$ for $\hat{D}_{\tilde{j}}^+$ or $q \leq \tilde{N}$ for $\hat{D}_{\tilde{j}}^-$, and solving for $\tilde{j}$, we get,

$$\frac{E + \ell}{2} = J_0^{3(\text{discrete})} = q + w + \frac{1}{2} + \sqrt{\frac{1}{4} + (k-2) \left(\tilde{N} - wq + h - 1 - \frac{1}{2}w(w+1) \right)} \quad (72)$$

The bound, $\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2}$, for the discrete representations, translates to,

$$\frac{k}{4}w^2 + \frac{w}{2} \leq \frac{1}{4(k-2)} + \tilde{N} - wq + h - 1 \leq \frac{k}{4}(w+1)^2 - \frac{(w+1)}{2} \quad (73)$$

For the continuous representation, $\tilde{j} = 1/2 + is$, we get,

$$\frac{E + \ell}{2} = J_0^{3(\text{continuous})} = \frac{k}{4}w + \frac{1}{w} \left(\frac{\frac{1}{4} + s^2}{k-2} + \tilde{N} + h - 1 \right) \quad (74)$$

A key result is that the discrete and continuous spectra join smoothly at the endpoints of the allowed range of $\tilde{j}$. At $\tilde{j} = \frac{1}{2}$, the discrete representation $\hat{\mathcal{D}}_{\tilde{j}}^{\pm, w}$ becomes degenerate with the continuous representation $\hat{\mathcal{C}}_{s=0}^{\alpha=\frac{1}{2}, w}$. Similarly, using $\hat{\mathcal{D}}_{\tilde{j}}^{+, w} \simeq \hat{\mathcal{D}}_{\frac{k-1}{2}-\tilde{j}}^{-, w+1}$, the upper endpoint $\tilde{j} = \frac{k-1}{2}$ also merges with an $s = 0$ continuous representation $\hat{\mathcal{C}}_{s=0}^{\alpha=\frac{1}{2}, w+1}$ in the $(w+1)$ -th flowed sector. Thus, short-string states described by discrete representations continuously connect to long-string states described by continuous representations.

For fixed h and w , the lowest-energy physical state is obtained by minimizing J_0^3 . The minimum of the discrete spectrum occurs at $q = \tilde{N} = 0$ and is lower than the continuous-spectrum minimum within the given range of h . At the boundaries of this range, the discrete and continuous ground states coincide, while outside it, the continuous representations provide the true ground state. Classically, the transition between short and long strings occurs at $h_{\text{crit}} = \frac{k}{4}w^2$. But, in the quantum theory, this transition broadens into a finite interval with a smooth crossover region.

Physical closed-string states must have identical left- and right-moving Casimir eigenvalues ($C_2 \sim$ string spacetime mass) and equal spectral flow, $w_L = w_R \equiv w$ (to satisfy locality and level matching). So, the complete AdS₃ string spectrum is,

$$\mathcal{H}_{\text{SL}(2, \mathbb{R})} = \bigoplus_{w \in \mathbb{Z}} \left(\bigoplus_{1/2 < \tilde{j} < (k-1)/2} \left(\hat{\mathcal{D}}_{\tilde{j}, L}^{\pm, w} \otimes \hat{\mathcal{D}}_{\tilde{j}, R}^{\pm, w} \right) \oplus \int ds \hat{\mathcal{C}}_{\frac{1}{2}+is, L}^{\alpha, w} \otimes \hat{\mathcal{C}}_{\frac{1}{2}+is, R}^{\alpha, w} \right) \quad (75)$$

4 Scattering of Long Strings and Reflection Coefficients

Long strings can reach the boundary of AdS_3 , so their dynamics are naturally described in terms of scattering processes. After Wick rotating both the worldsheet and target-space time coordinates, Lorentzian AdS_3 is analytically continued to the Euclidean space $H_3^+ \cong \text{SL}(2, \mathbb{C})/\text{SU}(2)$, described by the H_3^+ WZW model. Here, $\rho \rightarrow \infty$ becomes the conformal boundary of H_3^+ with a geometry of the 2-dim. Euclidean cylinder, where vertex operators are inserted and scattering amplitudes are obtained from boundary correlation functions. In this description, the worldsheet coordinates become holomorphic and anti-holomorphic, the affine currents admit Laurent expansions, and the Hermiticity conditions $(J_n^3)^\dagger = J_{-n}^3$ and $(J_n^\pm)^\dagger = J_{-n}^\mp$ remain inherited from the Lorentzian theory. So, the H_3^+ WZW model preserves the same $\widehat{\mathfrak{sl}(2, \mathbb{R})}_{k_L} \times \widehat{\mathfrak{sl}(2, \mathbb{R})}_{k_R}$ affine symmetry structure.

The Cartan subalgebra of $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ forms an affine $\widehat{\mathfrak{u}(1)}_k$ current algebra and hence can be represented by a free timelike boson ϕ . Matching the free-boson two-point function with the Cartan-current OPE gives the bosonization,

$$J_R^3(z) = -i\sqrt{\frac{k}{2}} \partial\phi(z) \quad , \quad J_L^3(\bar{z}) = -i\sqrt{\frac{k}{2}} \bar{\partial}\phi(\bar{z}) \quad (76)$$

which reproduces the Cartan current algebra. The ladder currents can be written as,

$$J_R^\pm(z) = \psi^\pm(z) e^{\pm i\sqrt{\frac{2}{k}}\phi(z)} \quad , \quad J_L^\pm(\bar{z}) = \bar{\psi}^\pm(\bar{z}) e^{\pm i\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})} \quad (77)$$

where $\psi^\pm$ and $\bar{\psi}^\pm$ fields carry the non- $U(1)$ degrees of freedom. These fields are neutral under the Cartan current and are chosen so that the full affine $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ algebra is reproduced. The total stress tensor decomposes into a free-boson and parafermion contribution, $T = T^{(J^3)} + T_{\text{PF}}$, $\bar{T} = \bar{T}^{(J^3)} + \bar{T}_{\text{PF}}$ corresponding to the decomposition $\text{SL}(2, \mathbb{R}) \sim \text{SL}(2, \mathbb{R})/U(1) \times U(1)$. Requiring the currents $J^\pm$ to have conformal weight of ‘one’, fixes the conformal weights of $\psi^\pm$, $\bar{\psi}^\pm$ fields to $h_\psi = \bar{h}_{\bar{\psi}} = 1 + \frac{1}{k}$. The states in the unflowed representation correspond to the primary operators,

$$\Phi_{m\bar{m}}^j(z, \bar{z}) = :e^{im\sqrt{\frac{2}{k}}\phi(z)+i\bar{m}\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \Psi_{m\bar{m}}^j(z, \bar{z}) \quad (78)$$

where $\Psi_{m\bar{m}}^j$ is called the Parafermion field which is neutral under $J_{R,L}^3$. Its conformal weights split into a Cartan part and a parafermion part,

$$h_\Phi^{(J^3)} = -\frac{m^2}{k} \quad , \quad h_\Phi^{(PF)} = h_\Psi = -\frac{j(j-1)}{k-2} + \frac{m^2}{k} \quad (79)$$

with the same expressions for the anti-holomorphic sector. Under spectral flow by an integer w , the bosonized primary field is replaced by

$$\Phi_{\tilde{m}\tilde{\bar{m}}}^{\tilde{j},w}(z, \bar{z}) = :e^{i(\tilde{m}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\phi(z)+i(\tilde{\bar{m}}+\frac{k}{2}w)\sqrt{\frac{2}{k}}\bar{\phi}(\bar{z})}: \Psi_{\tilde{m}\tilde{\bar{m}}}^{\tilde{j}}(z, \bar{z}) \quad (80)$$

This changes the eigenvalues and conformal weights of the primary field as dictated by the representation-theoretic definition of spectral flow.

$$m = \tilde{m} + \frac{k}{2}w \quad \text{and} \quad h_\Phi = \frac{-\tilde{j}(\tilde{j}-1)}{k-2} - w\tilde{m} - \frac{k}{4}w^2 \quad (81)$$

Long-string states belong to the spectrally flowed continuous representations $\hat{\mathcal{C}}_{j=\frac{1}{2}+is}^{\alpha,w}$. For $\tilde{N} = 0$, their vertex operators are labelled by the quantum numbers $(j, \tilde{m}, \tilde{\bar{m}}, w)$, with spectral flow shifting the charges as $m = \tilde{m} + \frac{k}{2}w$ and $\bar{m} = \tilde{\bar{m}} + \frac{k}{2}w$. The spacetime energy and angular momentum are $E = m + \bar{m}$ and $\ell = m - \bar{m}$, while s plays the role of the radial momentum since the asymptotic wavefunctions behave as $e^{\pm is\rho}$. Imposing the physical-state conditions $L_0 + h = \bar{L}_0 + \bar{h} = 1$ determines $\tilde{m}$ and $\tilde{\bar{m}}$ in terms of w , s , and the internal conformal weights. Since, $j = \frac{1}{2} + is$ and $1 - j = \frac{1}{2} - is$ describe the same physical state, corresponding respectively to outgoing and incoming radial waves $e^{\pm is\rho}$, one has the reflection relation, $\Phi_j = R(j, \tilde{m}, \tilde{\bar{m}}) \Phi_{1-j}$, where R is the reflection coefficient. So, the exact two-point function for the physical string states can be written as,

$$\left\langle \Phi_{m\bar{m}}^j(z, \bar{z}) \Phi_{m'\bar{m}'}^{j'}(z', \bar{z}') \right\rangle = R(j, \tilde{m}, \tilde{\bar{m}}) \delta(s - s') \delta(E + E') \delta_{N+N'} \frac{1}{|z - z'|^4} \quad (82)$$

Orthogonality of scattering states gives the factor $\delta(s - s')$. Likewise, $\delta(E + E')$ and $\delta_{N+N'}$ enforce conservation of spacetime energy and oscillator level, respectively. The coordinate dependence is completely fixed by worldsheet conformal invariance; since physical vertex operators satisfy $h_{\text{tot}} = \bar{h}_{\text{tot}} = 1$. The remaining dynamical information is contained in the reflection coefficient $R(j, \tilde{m}, \tilde{\bar{m}})$, which measures the amplitude for an incoming long string to be reflected to the asymptotic region. For the continuous representations with $j = \frac{1}{2} + is$, the reflection coefficient takes the form

$$R(s, \tilde{m}, \tilde{\bar{m}}) = \frac{\Gamma(\frac{1}{2} + is - \tilde{m}) \Gamma(\frac{1}{2} + is + \tilde{\bar{m}}) \Gamma(-2is) \Gamma(\frac{2is}{k-2})}{\Gamma(\frac{1}{2} - is - \tilde{m}) \Gamma(\frac{1}{2} - is + \tilde{\bar{m}}) \Gamma(2is) \Gamma(-\frac{2is}{k-2})} \quad (83)$$

with the w -dependence encoded implicitly through $\tilde{m} = m - \frac{k}{2}w$ and $\tilde{\bar{m}} = \bar{m} - \frac{k}{2}w$. The reflection amplitude is a tree-level closed-string scattering amplitude computed from a two-point function on the sphere. After fixing conformal gauge, the residual worldsheet symmetry is the global conformal group $\text{PSL}(2, \mathbb{C}) \simeq \text{SL}(2, \mathbb{C})$. Fixing the two operator insertions at 0 and ∞ leaves a residual scaling symmetry whose divergent volume cancels the factor $\delta(0)$ arising from energy conservation, yielding a finite physical amplitude.

The poles of R encode the discrete bound states. The poles of $\Gamma(\frac{1}{2} + is - \tilde{m})$ occur at $\tilde{m} = \tilde{j} + q$ ($q = 0, 1, 2, \dots$), reproducing precisely the mass-shell condition of the discrete representations and hence the short-string spectrum. Poles of $\Gamma(\frac{1}{2} + is + \tilde{\bar{m}})$ similarly reproduce the highest-weight discrete series. However, not every pole corresponds to a physical state. Examining the residues shows that their sign changes at $\tilde{j} = \frac{k-1}{2}$, with positive residues corresponding to positive-norm states and negative residues to ghosts. Consequently, only poles satisfying $\frac{1}{2} \leq \tilde{j} \leq \frac{k-1}{2}$ represent physical discrete states. Thus, the pole structure of the long-string reflection amplitude reproduces both the discrete spectrum and the no-ghost bound from a scattering-theoretic perspective.

Conclusion

In this report, we studied bosonic string theory in $\text{AdS}_3 \times \mathcal{M}$ using the exact worldsheet description provided by the $\text{SL}(2, \mathbb{R})$ WZW model. Beginning with the realization of AdS_3 as the group manifold of $\text{SL}(2, \mathbb{R})$, we constructed the classical theory, derived its chiral current algebra and Virasoro constraints, and analyzed geodesic solutions. We showed that spectral flow generates new classical solutions corresponding to winding strings, giving rise to short-string and long-string configurations.

Upon quantization, the conserved currents generate the affine $\widehat{\mathfrak{sl}(2, \mathbb{R})}_k$ algebra, while the Sugawara construction produces the corresponding Virasoro algebra. We studied the unitary, discrete, and continuous representations of both the global and affine algebras and established the no-ghost theorem, demonstrating that physical states are ghost-free after imposing the Virasoro constraints. Spectral flow was then shown to be an automorphism of the spectrum-generating algebra, making the inclusion of spectrally flowed representations essential for a complete description of the physical Hilbert space. The physical spectrum consists of spectrally flowed discrete representations describing short strings and spectrally flowed continuous representations describing long strings, with a smooth transition between the two sectors at the boundaries of the allowed discrete spectrum.

Finally, we investigated long-string scattering by analytically continuing the theory to the Euclidean H_3^+ WZW model. Using bosonization and parafermionic techniques, the primary fields were constructed, and the reflection amplitude was analyzed. The poles of the reflection coefficient were shown to correspond to normalizable discrete states, while their residues reproduce the unitarity bound required by the no-ghost theorem. Thus, long-string scattering provides an independent and complementary derivation of the physical spectrum, establishing a direct connection between the affine representation theory and the analytic structure of scattering amplitudes in string theory in AdS_3 .

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